

A Survey on Educational Data Mining [2014-2019]

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Abstract—Nowadays Data Mining is used in many application areas enabling large data streams and algorithms for analysis and extraction of powerful data. On their side, the Computer Environments for Human Learning (EIAH) offer TEL devices (Technology-enhanced learning) such as simulators, serious games, MOOCs (massive online open courses), or educational platforms. These devices provide data that are traces of the activities of students or teachers. The data produced are cognitive information of very fine levels (student knowledge, skills, and errors) and require specific analysis and processing tools, we talk here about educational data mining methods, Educational data processing (EDM) is rising as a notion of research and analysis with a set of machine and psychological ways and research approaches for understanding however students learn. EDM uses machine approaches to research instructional knowledge so as to review instructional queries. For this knowledge exploration, several tools were used like personal learning environments, recommender systems, Context learning, and Course management systems. These tools offer numerous edges for instructional data processing. In this survey, we have a tendency to focus and supply numerous tools of analysis trends exploitation EDM Tools to explore data and knowledge, and explaining the process of EDM application, the goal is not only to transform the data into knowledge but also to filter the extracted knowledge to know how to modify the educational environment to improve learners' learning. This paper surveys the foremost relevant studies administrated during this field up to date.

Keywords: Data mining, educational data mining, EIAH, E-learning.

I. Introduction:

Data mining is a collection of methods that extract information (semi-automatically) from large databases. The information discovered is rules, associations, unknown trends (not fixed in advance), and special structures. They must help to make decisions.

The term data mining is now used in a large number of applications of all types in various fields: Healthcare and Insurance, Financial Data Analysis, Telecommunication, and specially Education.

Educational data mining (EDM) has become an independent analysis space in recent years With group of machine and psychological

strategies and analysis approaches for understanding however students learn, EDM develops strategies and applies techniques from machine learning, statistics, and data processing to research information collected throughout teaching and learning [1]. In this work, we tend to hold out a survey on recent educational data mining. We will present in detail in a first part definition of data mining and its different fields of application, then we will present in a detailed way the methods used recently in the EDM and at the last part of this work, we will interpret the results to choose the most data mining methods used in education.

The term data mining is a process of discovering new correlations, characteristics, and trends by sifting through large amounts of data stored in deposits. This operation requires the use of forms recognition technologies, Statistics, and mathematical techniques, in another way; it is a set of techniques that aim to extract knowledge from large data. Thus, it consists of a family of tools, whether automatic or semi-automatic allowing the analysis of a large quantity of data contained in a database, for a purpose to identify recurrent patterns and to establish relationships of Correlation.

Data Mining also called Knowledge Discovery in Databases (KDD) is the field of discovering novel and potentially useful information from large amounts of data. Data mining has been applied in a great number of fields, including retail sales, bioinformatics, and counter-terrorism. [2].

In the field of education, educational data mining (also referred to as “EDM”) is a recent research area that explores and analyzes the information stored in student databases in order to understand and improve the performance of the student learning process. [3], also to design models, tasks, methods, and algorithms for exploring data from educational settings. EDM pursues to find out patterns and make predictions that characterize learners’ behaviors and achievements, domain knowledge content, assessments, educational functionalities, and applications. Source information is stored in repositories managed by conventional, open, and distance educational modalities. [4]

The present work extends the period described by earlier surveys, summarized in Section “number of section”, that cover from 2014 up to 2016. The aim is to preserve and update the chronicles of recent EDM development. The scope of the work is limited and provides a partial image of the EDM activity published in all celebrated events and available media.

II. Data Mining Applications

The fields of application of Data mining are very diverse we quote the most essential: health, Telecoms, Financial and data analysis, education and E-learning.



Figure 1 : Data Mining Applications

We will treat in this article the two following fields of applications considering their importance and their excellent effect: E-learning and Education.

A. E-learning:

In this section, we choose to present major applications of data mining in the field of e-learning and we present them as follow:

- Assessment of student learning :

The first observation was to examine how learners spend their time with materials and of course time spent with each page [3]. Therefore, a better understanding of how to evaluate the effectiveness of the development and delivery of courses in this way, such as the extraction of data from the solution has been used to provide ways to learn about learners through statistical analysis in order to make effective decisions regarding teaching methods.

For the same purpose (statistical analysis) Data Mining has been applied by basic statistical methods so that they can develop an integrated e-learning environment that includes learning history and learning content information to control the Training of each learner [5].

As well as the quality of the test that was related to how the exam sheet was constituted by meeting the multiple evaluation requirements [6], such as average degree of difficulty, degree of discrimination Average, the duration of the test time and its element numbers ..., To solve this problem, a test generator with multiple criteria is formulated and a dynamic programming approach is proposed. The proposed new approach uses data extraction, in particular grouping techniques, dynamic programming and fuzzy logic theory, which allow the grouping of test sheets according to the specified requirements.

- Evaluation of adaptation policies:

The adaptation of human learning systems (HLS) is one of the main obstacles to their development. The challenge in this area is to satisfy learners, or a group of learners, who use an HLS by understanding their behaviors, although most of the web-based educational systems are usually presented statically, Learners' goals, experiences, current knowledge and abilities , and without interactivity means there are fewer opportunities for instant answers and feedback Of the instructor when online learners need support .To personalize and adapt learning can ensure that the system takes into account the strengths and weaknesses of each individual Which uses the program this realization leads to the reuse of the profiles of the learners in order to be able to evaluate the systems of assignment of the courses [7]. This re-use is based on the results obtained from the evaluation for each course to evaluate the policy applied in the adaptation phase of learning for the learners before processing the results obtained by these learners. The results must satisfy two conditions: the success rate and the level of competence between learners [8].

- Interest and added value of data mining techniques in the field of e-learning :

The system exploits Data Mining techniques to solve certain problems:

Customizing e-learning, recommending learning activities and tasks, suggesting interesting learning experiences for students, shortening path or a simple link to follow, and generating adaptation tips that can be used by students or learners [9].

Obtaining objective information about teaching, learning analysis and student behavior, detecting students in need of support, predicting student achievement, classifying learners into groups, Regular and irregular patterns of a learner, detection of the most frequent mistakes made, determination of the most effective activities and improvement of adaptation and customization of courses are solved problems for trainers, teachers, Instructors or tutors [10].

Improved adaptation and customization of student learning, assessment of the structure of course content and its effectiveness in the learning process, automatically build student models and models of tutors, comparison Data mining techniques in order to be able to recommend the

most useful for each task and the development of specific data mining tools for educational purposes that may be beneficial Course developers and researchers in education [10].

The improvement of the decision-making process in higher education institutions, the rationalization of efficiency in the decision-making process, the attainment of specific objectives, the suggestion of certain courses which could be useful for each category of education, Learners and admission assistance for students who will do well in universities that can be used by organizations, learning providers, universities and private training companies [10].

The development of the best way to organize the institutional (human and material) resources and their educational offer, the use of available resources more effectively, the improvement of educational programs and the effectiveness of the Distance learning, teacher evaluation and curriculum development, and setting parameters to improve website efficiency and adapt it to users (optimal server size, distribution of network traffic, etc.) Can be beneficial to administrators, school administrators, and administrators of networks [10].

B. Education:

A new domain exists, called Educational Data mining, which aims to analyze and process the developmental methods used in teaching and education, it adds an option to predict the future behavior of students in the classroom. Learning, studying the effects of pedagogical support and the advancement of scientific knowledge about learning, data mining can be used by an institution for decision support and also to predict student outcomes.

With the results, the institution concerned can afterwards focus on what to teach and how to teach it.

III. Educational Data mining:

A first definition says that EDM is a discipline that relies on the techniques of data analysis applied to educational data, It is a method aimed at improving the quality of teaching and learning, Developing specific methods for the Exploring data produced from educational contexts and using them to better understand students and the contexts in which they learn.[11]

Another considers EMD mutually of the foremost appropriate technology in providing new knowledge about the behavior of lecturer, student, alumni, manager, and other educational staff and acting as an energetic automated assistant in helping them to form better decisions on their educational activities.[12], according to these definitions we come to the conclusion that there are several methods and tools developed for educational data mining.

A. Data mining techniques in educational systems:

In this section, we'll gift very well the various data processing techniques in education and learning, Currently, there are several methods of educational data mining that are extracted from various literatures, especially data mining and machine learning, psychometrics and other areas of computer modeling, information visualization statistics and computer modeling [14], as a first technique we will focus on prediction.

• Prediction :

It is a data mining technique that allows us to identify a variable based simply on the description of other related and antecedent variables, we distinguish several types of this technique we quote for example: Classification, regression.

Another point of view [11] says that the prediction it is a model of numerical data to predict unknown or missing values, a technique that predicts a future state rather than a current state. It is used in the field of education to predict student performance, success rate, their knowledge, score, or marks; In this case, we estimate the unknown value of a variable that describes the scholar. This value will be numerical/continuous (regression task) or categorical/discrete (classification task), Regression analysis is employed to search out relation between a variable quantity and one or more independent variables.

Classification

is employed to cluster individual things primarily based upon quantitative characteristics inherent within the things or on coaching set of antecedently labeled things.[15], in addition a description that goes in the same direction [16], define the prediction in educational data mining as a technique used to predict the performance of the student. In order to predict the performances of the student, several tasks in are used which is classification, regression. **Classification** is a supervised learning technique whose aim is to create a model, in this specific case, called classifier, which can classify the class label of unknown data, **Regression** analysis is a statistical methodology that is most often used in numeric prediction. The objective of this task is to achieve a function of the independent variables that allows computing the conditional expectation of a dependent variable for prediction and forecasting exercises based on the minimization of a certain type of error via an iterative procedure. [16]

A second educational data mining technique is clustering: is based on the principle of maximizing intragroup similarity and minimizing it between distinct groups, also it can offer helpful outcomes and results that assist in addressing several problems and issues within the instructional domain. [17]

Clustering is the most method applicable to educational data mining to better identify teams of comparable observations, or Grouping similar materials or

students supported their learning and interaction patterns.[13]

The Clustering, it can support administrative and technical activities in higher education; apply this technique to generate models of future recommendations for students who have similar performances. [11]

In other words The Clustering [11] is Could be a technique accustomed separate knowledge into completely different teams supported sure common options and offers the user a broad read of what's happening in this dataset. [14]

The third educational data mining technique is **Relationship mining (RM)**, **Relationship mining** is a solution that aims to help universities build techniques and policies to reduce the drop-out rate of students or increase the retention and success rate, to prove better results learning.

RM is a study of relationships among variables and to encode rules. Identifying relationships in learner behavior patterns and diagnosing student difficulties. [13]

The main mission of relationship mining is to check whether an event causes another event or not by doing analysis on the coverage of the two events in the data set. [14]

Another area that is part of educational data mining technique is Distillation of data for human judgment; its main goal is to make complex data understandable by humans to leverage their judgment, in many situations, humans can make interpretations about the data when it presented correctly.

Distillation of data for human judgment in general context is to represent data in intelligible ways using summarization, visualization and interactive interfaces, in education is Helping instructors to visualize and analyze the ongoing activities of the students and the use of information. [13]

Distillation of data for human judgment there are two objectives for human judgment for which distillation of data can be done: Identification and Classification, in this case data is represented in a way that human can easily recognize the well specified patterns.[15]

Distillation of data for human judgment The objective of this method is to present data in summarize and visualized way for human, in order to focus on appropriate information and support decision making . [13] The last technique is

Discovery with models (DwM):
In the Discovery with a Model method, a model is developed via prediction, clustering or by human reasoning knowledge engineering and then used as a component in another analysis, specifically in prediction and relationship mining.

DwM: [13] Identification of relationships among student behaviors and characteristics or contextual variables.

Integration of psychometric modeling frameworks into machine-learning models, in order to employ a previously validated model of a phenomenon as a component in another analysis, another point of view describes **DwM** technique as a model which generally supported by clustering, prediction, or knowledge engineering using human reasoning rather than automated methods. The developed model is then used as a part of different comprehensive models like relationship mining. [14]

B. The EDM application process:

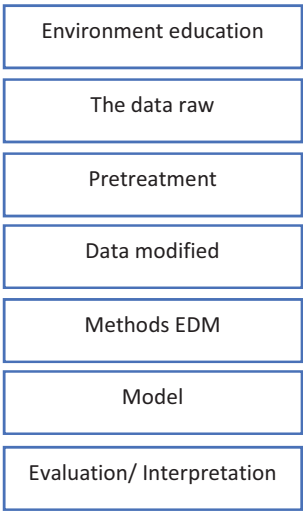


Figure 2: The data mining application process applied in education [20].

The data mining application process applied in education can be considered as an iterative cycle of hypothesis formulation, testing, and refinement [18]. In this process, the goal is not only to transform the data into knowledge but also to filter the extracted knowledge to know how to modify the educational environment to improve learners' learning (figure 2).

Romero and Ventura [18] and Sachin and Vijay [19] found that the EDM application is similar to the Knowledge Extraction from Data process. This process begins with collecting data for use from the educational environment. The raw data that has been obtained requires cleaning and preprocessing such as: Merging heterogeneous data, processing missing data, converting data, etc. This phase often requires the use of certain data mining techniques.

The result of the latter is a model that can structure the data that is stored. Finally, the last step is the interpretation and evaluation of the results obtained.

C. Challenges of educational data mining:

- Most data processing techniques are unit engineered for business knowledge and extract business worth.
- The sequence represents however ideas depend on each other and guide however apply and tutoring ought to be ordered.
- Context is important for explaining results and knowing where a model may or may not work.
- Building community in learning spaces in the face of the current emphasis on scaling up and personalization.
- Keeping formal education relevant in a world where information is everywhere .
- Facilitating and assessing learning trajectories (not momentary states).
- Adapting to increasingly diverse learner populations online and face-to-face.

D. Goals of educational data mining:

- Predict future student performance (based on past patterns of learning across numerous student bodies).
- Personalize the training method for every and each student, enjoying to their strengths and inspiring improvement.
- Adapt teaching and learning designs via socialization, pedagogy, and technology.
- Study the consequences of the effects of educational support through learning systems.
- Advancing scientific knowledge about the learning environment and learners - By building models appropriate to the student.

E. Educational data mining references:

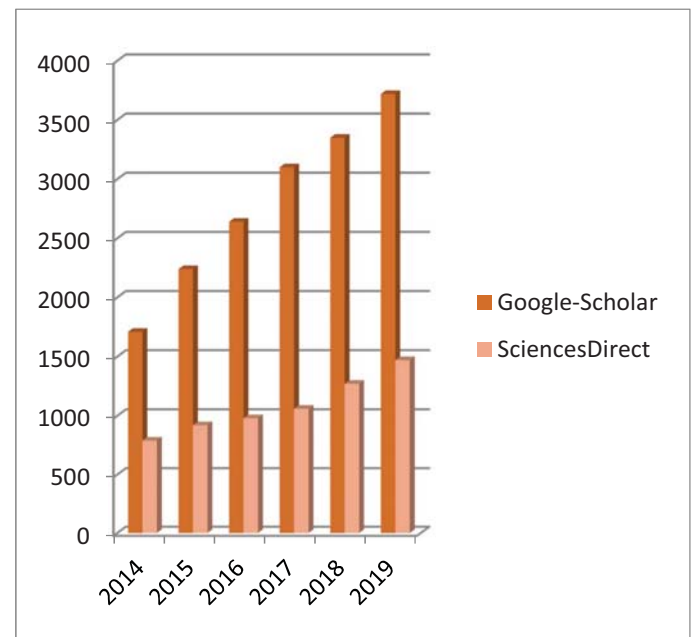


Figure 3 : Number of educational data mining references in Google Scholar and science Direct by year.

The figure 3 shows the number of scientific studies related to educational data mining that corresponds to the results of a research from 2014 to 2019 in Google scholar and Science Direct. As we can see, the numbers are increasing, showing great interest in this area of research.

IV. Conclusion and interpretation:

Educational data mining is one of the new trends and techniques that play a crucial role in improving the quality of higher education by developing data mining methods from educational contexts, and use these methods to better understand students and their learning environments, From the results of our research, we come to detect that the most widely used data mining techniques in the field of education and higher education are: prediction, clustering, relationship mining, distillation of data for human judgment and discovery with models.

Data mining methods have contributed to the development of learning processes and models, for teacher and student to help them to accomplish their assignments. These models come from classification methods, regression methods, clustering methods, and relationship mining methods distillation of data for human judgment methods

and discovery with model methods, and all of that to achieve an improvement of conditions in learning environments.

In this study, we highlighted scientific works and studies related to the Data Mining, area application of data mining and educational data mining, Interest in the use of data mining in the field of e-learning. Current research efforts in this area focus on exploring the performance of data mining in solving new problems related to intelligent learning system user interactions, after the analysis of EDM techniques and by treating all the references found and illustrated at the level of the part Educational data mining references, prediction, clustering, relationship mining, distillation of data for human judgment and discovery with models are techniques that show immense effectiveness in the field of education .

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